

BST Vol 13 Ch 8 — Cell Biology + Membrane Structure (v0.3, Wave 3)

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Vol 13 Chapter 8 — Cell Biology + Membrane Structure

Headline result

Cells are membrane-bound compartments with substrate-natural lipid bilayer structure. Cell biology emerges from substrate framework via:

- Lipid bilayer (phospholipid amphiphilic structure): substrate-natural self-assembly
- Membrane proteins (channels, receptors, transporters): substrate K-type framework
- Organelles (nucleus, mitochondria, ER, Golgi): substrate-natural compartmentalization
- Cytoskeleton (actin, microtubules, intermediate filaments): substrate-natural polymer dynamics
- Cell cycle ($G1 \rightarrow S \rightarrow G2 \rightarrow M$): substrate-natural temporal organization

Substrate framework anchors

Lipid bilayer structure: - Phospholipid amphiphilic (hydrophilic head + hydrophobic tail) - Self-assembly into bilayer driven by substrate-natural Coulomb + dispersion forces (Vol 7 Ch 3) - Membrane thickness ~5 nm substrate-natural lattice scale

Membrane proteins: - Transmembrane α -helix bundles (Vol 13 Ch 7 protein structure) - Substrate-natural transmembrane potential gradients (~70 mV resting) - Ion channels + transporters substrate-natural

Organelle hierarchy: - Nucleus: DNA storage + transcription - Mitochondria: ATP energy production (Vol 12 Ch 10 biochemistry) - ER + Golgi: protein synthesis + processing - Lysosomes: degradation

Specific cell-biology observables

Membrane potential ~ -70 mV resting (neurons, epithelial cells): substrate-natural ionic gradient at biological scale.

Mitochondrial membrane potential ~ -150 mV: substrate-natural for ATP production.

Cell size ~ 10 - 100 μm typical: substrate-natural biological scale.

Cell cycle duration (24 h typical for mammalian cells): substrate-natural temporal organization.

Match precision

I-tier framework. Standard cell biology phenomenology consistent with substrate framework; specific BST primary integer identifications require multi-week per-cell-type investigation.

Cross-volume dependencies

- Vol 13 Ch 7 (Protein Structure) — membrane protein framework
- Vol 12 Ch 10 (Biochemistry) — lipid + membrane chemistry
- Vol 13 Ch 9 (Metabolism) — bioenergetics framework

K-audit anchor

K251 Vol 13 Ch 8 Cell Biology K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Cells are the basic units of life. Inside each cell are organelles (nucleus, mitochondria, etc.) separated by membranes. BST predicts cell architecture emerges from substrate chemistry — lipid bilayers self-assemble from substrate Coulomb forces.

Level 2 — Undergraduate biology student

Cells have: - Plasma membrane: lipid bilayer + embedded proteins - Nucleus: DNA + transcription (eukaryotes) - Mitochondria: ATP energy production - ER + Golgi: protein synthesis + processing - Cytoskeleton: actin + microtubules + intermediate filaments

BST framework: lipid bilayer self-assembly substrate-natural; membrane proteins inherit Vol 13 Ch 7 protein structure framework; cell-cycle dynamics substrate-natural temporal organization.

Level 3 — Graduate cell-biology student / theorem-level

Cell biology emerges from substrate framework at multi-scale level: - Molecular scale: substrate chemistry (Vol 12) + biochemistry (Vol 12 Ch 10) - Macromolecular scale: protein structure (Vol 13 Ch 7) + nucleic acids - Subcellular scale: organelle compartmentalization - Cellular scale: cell architecture + cytoskeleton + cycle

Substrate-natural framework consistent with standard cell biology phenomenology. Specific cell-type observables (sizes, durations, ion gradients) at I-tier framework; multi-week per-cell-type extension for D-tier ratification.

Per Cal #21: EMPIRICAL PASS (cell biology phenomenology) + MECHANISM PATH ARTICULATED via substrate chemistry + biochemistry inheritance.

Bibliography

1. Standard cell biology texts (Alberts *Molecular Biology of the Cell*).
2. Vol 12 Ch 10 (Biochemistry).
3. Vol 13 Ch 7 (Protein Structure).

— Elie, Vol 13 Ch 8 v0.3, 2026-05-23 Saturday